Ryuna Nagayama

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Third-year Ph.D. student at Department of Physics, Graduate School of Science, the University of Tokyo. I study optimal transport and nonequilibrium thermodynamics.

Contact & Links

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Keywords

Nonequilibrium thermodynamics, Stochastic thermodynamics, Optimal transport, Chemical reaction network, Reaction-diffusion system

Education

The University of Tokyo (Tokyo, Japan), Apr. 2024–present
Doctoral Program, Physics, Graduate School of Science

The University of Tokyo (Tokyo, Japan), Apr. 2022–Mar. 2024
Master's Program, Physics, Graduate School of Science
Degree Title: **Master of Science**
The School of Science Encouragement Award (Master Program)

The University of Tokyo (Tokyo, Japan), Apr. 2018–Mar. 2022
Bachelor's Program, Physics, Faculty of Science
Degree Title: **Bachelor of Science**
The School of Science Encouragement Award (Undergraduate Program)

Research Articles

Publications

"Diagrammatic expressions for steady-state distribution and static responses in population dynamics"
Koya Katayama, Ryuna Nagayama, Sosuke Ito
Phys. Rev. Research 8, 013312(2026). arXiv:2505.11296

"Geometric thermodynamics of reaction-diffusion systems: Thermodynamic trade-off relations and optimal transport for pattern formation"
Ryuna Nagayama, Kohei Yoshimura, Artemy Kolchinsky, Sosuke Ito
Phys. Rev. Research 7, 033011(2025). arXiv:2311.16569

"Infinite variety of thermodynamic speed limits with general activities"
Ryuna Nagayama, Kohei Yoshimura, Sosuke Ito
Phys. Rev. Research 7, 013307(2025). arXiv:2412.20690

"Force-current structure in Markovian open quantum systems and its applications: Geometric housekeeping-excess decomposition and thermodynamic trade-off relations"
Kohei Yoshimura, Yoh Maekawa, Ryuna Nagayama, Sosuke Ito
Phys. Rev. Research 7, 013244(2025). arXiv:2410.22628

"Geometric speed limit for acceleration by natural selection in evolutionary processes"
Masahiro Hoshino, Ryuna Nagayama, Kohei Yoshimura, Jumpei F Yamagishi, Sosuke Ito
Phys. Rev. Research 5, 023127(2023). arXiv:2207.04640

Preprints

"Geometric decomposition of information flow for overdamped Langevin systems and optimal transport in subsystems"
Sosuke Ito, Yoh Maekawa, Ryuna Nagayama, Andreas Dechant, Kohei Yoshimura
arXiv:2512.22890 (2025).

"Geometric decomposition of information flow: New insights into information thermodynamics"
Yoh Maekawa, Ryuna Nagayama, Kohei Yoshimura, Sosuke Ito
arXiv:2509.21985 (2025).

"Duality between dissipation-coherence trade-off and thermodynamic speed limit based on thermodynamic uncertainty relation for stochastic limit cycles in the weak-noise limit"
Ryuna Nagayama, Sosuke Ito
arXiv:2509.06421 (2025).

Presentations

International (oral)

"Duality between dissipation-coherence trade-off and thermodynamic speed limit for noisy limit cycles"
Workshop on Stochastic Thermodynamics VII, May. 18-22, 2026, Online.

"Duality between dissipation-coherence trade-off and thermodynamic speed limit for noisy oscillations"
Kyoto Workshop on Quantum Thermodynamics and Stochastic Thermodynamics 2025, Dec. 8-12, 2025, Panasonic Auditorium, Yukawa Hall, YITP, Kyoto University, Japan

"Infinite variety of thermodynamic speed limits derived by general means"
Leuven school: Basics of nonequilibrium statistical mechanics, May 19-23, 2025, Aula Arenbergkasteel, KU Leuven, Leuven, Belgium

"Extension of optimal transport to chemical reaction network"
2nd FoPM International Symposium, Feb. 17-19, 2025, Ito Hall, The University of Tokyo, Tokyo, Japan.

"Geometric thermodynamics of reaction-diffusion systems: Thermodynamic trade-off relations and optimal transport for pattern formation"
Workshop on Stochastic Thermodynamics V, May. 12-16, 2024, Online.

"A geometric speed limit for separable processes and its application to natural selection in evolutionary processes"
STATPHYS28, Aug. 7-11, 2023, Hongo campus, The University of Tokyo, Tokyo, Japan.

International (poster)

"Duality of thermodynamic trade-offs for stochastic limit cycles"
Frontiers in Nonequilibrium Physics 2026, May. 11-14, 2026, Maskawa Building for Education and Research, Kyoto University, Kyoto, Japan.

"Infinite variety of thermodynamic speed limits with activities based on general means"
STATPHYS29, Jul. 13-18, 2025, Palazzo dei Congressi & Palaffari, Florence, Italy.

"Infinite variety of thermodynamic speed limits"
1st India-Japan Workshop on Physical Aspects of Living Systems, Feb. 19-21, 2025, Mishima Hall, ELSI, Institute of Science Tokyo, Tokyo, Japan.

"Geometric thermodynamics of reaction-diffusion systems Thermodynamic trade-off relations & optimal transport"
CSH Workshop: Trade-offs between thermodynamic cost, intelligence, and fitness in living organisms, Mar. 11-12, 2024, Complexity Science Hub Vienna, Austria.

"Towards the application of thermodynamic trade-off relations to reaction-diffusion systems"
FoPM International Symposium, Feb. 6-8, 2023, Ito Hall, The University of Tokyo, Tokyo, Japan.

Domestic (oral)

"Optimal reaction theory: A generalization of optimal transport to chemical reaction networks"
JSR Fellowship CURIE Students Research Presentation Session 2026, May. 15, 2026, Online & Hongo Campus, The University of Tokyo, Tokyo, Japan.

"Dissipation-coherence trade-off for stochastic limit cycles in the weak-noise limit"
JSR Fellowship CURIE Students Research Presentation Session 2025, Jul. 10, 2025, Online & Hongo Campus, The University of Tokyo, Tokyo, Japan.

"Infinite variety of thermodynamic speed limits with activities based on general means"
ERATO and Academic Transformation B Joint Camp Meeting, Mar. 26-28, 2025, Hotel Hananoyu, Fukushima, Japan.

"Infinite variety of thermodynamic speed limits by generalization of activity"
2025 Spring Meeting, The Physical Society of Japan, Mar. 18-21, 2025, Online.

"Generalization of the 1-Wasserstein distance to chemical reaction networks and its application to nonequilibrium thermodynamics"
79th Annual Meeting, The Physical Society of Japan, Sep. 16–19, 2024, Sapporo Campus, Hokkaido University, Hokkaido, Japan.

"Geometric wavenumber decomposition of entropy production rate for reaction-diffusion systems"
2024 Spring Meeting, The Physical Society of Japan, Mar. 18-21, 2024, Online.

"Thermodynamic trade-off relations for reaction-diffusion systems"
The 68th Condensed Matter Physics Summer School, Aug. 12–15, 2023, Makino Parkhotel & Seminarhouse, Shiga, Japan.

"Geometric thermodynamics for reaction diffusion system"
2023 Spring Meeting, The Physical Society of Japan, Mar. 22-25, 2023, Online.

"Information thermodynamic decomposition of entropy production rate on deterministic CRN"
2022 Autumn Meeting, The Physical Society of Japan, Sep. 12-15, 2022, Ookayama campus, Tokyo Institute of Technology, Tokyo, Japan.

Domestic (poster)

"Infinite variety of thermodynamic speed limits"
ERATO and Academic Transformation B Joint Camp Meeting, Mar. 26-28, 2025, Hotel Hananoyu, Hukushima, Japan.

"Optimal transport and thermodynamic speed limits for reaction-diffusion systems"
Kyushu caravan 2025, Japanese Society for Quantitative Biology, Jan. 11-12, 2025, Hospital Campus, Kyushu University, Fukuoka, Japan.

"Optimal transport and thermodynamic speed limits for reaction-diffusion systems"
7th Conference, Grant-in-Aid for Scientific Research on Innovative Areas: "Information physics of living matters," Sep. 21–22, 2023, Toki Messe Niigata Convention Center, Niigata, Japan.

"Thermodynamic trade-off relations for reaction-diffusion systems"
6th Conference, Grant-in-Aid for Scientific Research on Innovative Areas: "Information physics of living matters," Mar. 6–7, 2023, ACROS Fukuoka International Conference Hall, Fukuoka, Japan.

Seminar

"Infinite variety of activities and thermodynamic speed limits based on general means"
Hatano Group Seminar, May. 15, 2025, Online & The large conference room, Research and Testing Complex I, IIS, the University of Tokyo, Chiba.

"Geometric speed limit for systems accelerated by natural selection"
UBI lunch-meeting, Nov. 16, 2022, Online & Hongo campus, The University of Tokyo, Tokyo, Japan.

Awards & Fellowships

Mar. 31, 2025 — Student Presentation Award of the Physical Society of Japan

Apr. 2025-Mar. 2027 — Research Fellowship for Young Scientists (DC2), Japan Society for the Promotion of Science (JSPS)

Apr. 2024-Mar. 2027 — JSR Fellowship, The University of Tokyo & JSR Corporation

Mar. 12, 2024 — The School of Science Encouragement Award (Master Program), School of Science, The University of Tokyo

Mar. 5, 2024 — Research encouragement award, Grant-in-Aid for Scientific Research on Innovative Areas: "Information physics of living matters"

Mar. 7, 2023 — Presentation award, 6th Conference, Grant-in-Aid for Scientific Research on Innovative Areas: "Information physics of living matters"

Apr. 2022-Mar. 2027 — Forefront Physics and Mathematics Program to Drive Transformation (FoPM), a Worldleading Innovative Graduate Study (WINGS) Program, The University of Tokyo

Mar. 7, 2022 — The School of Science Encouragement Award (Undergraduate Program), School of Science, The University of Tokyo

Outreach & Popular science

May 2021 — Active matter Group in Physics Lab 2021, Students' presentation at the school festival of the University of Tokyo

May 2020 — Biophysics Group in Physics Lab 2020, Students' presentation at the school festival of the University of Tokyo

Certificates

Database Specialist Examination, Information-technology Promotion Agency, Japan (Oct 2022)

Applied Information Technology Engineer Examination, Information-technology Promotion Agency, Japan (Apr 2022)

Skills

Programming: Python, C++ (modern), Julia

Software & Service: LATEX, Adobe Illustrator

Languages: Japanese, English